

The Full Cycle of an ML Project

Training a model is just one part of a complete machine learning project. A successful system involves a full, iterative cycle that includes scoping, data, modeling, and deployment.

1. Scoping:

- This is the first step, where you **define the project and its goals**.
- *Example:* Deciding to build a speech recognition system for voice search.

2. Data:

- This involves **collecting, defining, and labeling** your dataset.
- *Example:* Gathering thousands of audio clips (x) and their corresponding text transcripts (y).

3. Modeling (Training):

- This is the iterative loop discussed in previous lessons:
 - Train the model.
 - Perform bias/variance and error analysis.
 - Use the insights to make decisions (e.g., get more data, change the architecture) and re-train.

4. Deployment and Monitoring:

- This is the process of **making your trained model available to users** in a "production" environment.
- You must continually **monitor** the model's performance to see if it's working well on real-world data.

The Process is an Iterative Loop

These steps are not a one-way street; they form a continuous cycle.

- **Loop 1 (Modeling $\leftrightarrow$ Data):** Error analysis (Step 3) might reveal your model is failing on audio with "car noise." This insight sends you back to Step 2 to **collect more specific data** (e.g., audio clips with car noise).
 - **Loop 2 (Deployment $\rightarrow$ Data):** Your live, deployed system (Step 4) will receive new data from users. This new data can be (with user permission) fed back into Step 2, allowing you to **continuously retrain and improve** your model.
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⚙️ Deployment in Practice: MLOps

"Deployment" is a significant engineering task that involves several key components:

- **Inference Server:** A dedicated server that hosts your trained model. Its sole job is to run the model and make predictions.
- **API Call:** Your application (e.g., a mobile app) sends the input data (like an audio clip, x) to the inference server via an API call. The server runs the model and returns the prediction (like the text transcript, $\hat{y}$).
- **System Monitoring & Maintenance:**
 - You must **log data and predictions** to monitor performance.
 - **Data can "drift"**: The real world changes. For example, new celebrity or politician names might appear that were not in your original training data, causing errors.
 - Monitoring allows you to catch this drop in performance and triggers a **model update** (i.e., you retrain the model with the new data).
- **MLOps (Machine Learning Operations):**
 - This is the name for the field and practice of systematically **building, deploying, and maintaining** machine learning systems.
 - It ensures your model is reliable, scalable, efficient (low compute cost), and can be updated as needed.